

C3MOD - CANCER CLUSTERING AND CHARACTERIZATION USING MULTIOMICS DATA

By

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Abstract

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Classification of patients with cancer is a crucial first step for clinicians to understand disease characteristics and manage effective treatments. C3MOD is a Python-based tool developed to advance cancer research by leveraging multiomics data for cancer subtype identification and characterization. The tool employs unsupervised machine learning algorithms, including KMeans, Similarity Network Fusion (SNF), and hierarchical clustering, to stratify patients based on datasets including miRNA, mRNA, and DNA methylation profiles. Its customizable framework allows researchers to analyze various cancer types, providing insights into survival outcomes, mutation burden, and immune characterization of identified subtypes.

C3MOD addresses the challenges in multiomics analysis, such as data integration and flexibility, by offering a user-friendly and scalable solution. It provides an efficient and accurate means of identifying cancer subtypes, which can facilitate precision medicine strategies tailored to individual patient profiles.

The tool has been validated using TCGA datasets for 30 cancer types, demonstrating its robustness and utility in identifying meaningful subtypes and biomarkers, such as GNG4 in lung

adenocarcinoma (TCGA-LUAD). This biomarker has been linked to poor survival outcomes, further underscoring the clinical relevance of C3MOD.

The open-source deployment of C3MOD on GitHub (<https://github.com/rajulshakya777/C3MOD>) ensures broad accessibility, empowering researchers to perform customized analyses and contribute to the growing field of personalized medicine. By integrating advanced clustering techniques with comprehensive downstream analyses, C3MOD establishes itself as a versatile and impactful tool for multiomics-driven cancer research.

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1. Introduction

The effective management and treatment of diverse diseases, including cancer, depend significantly on patient stratification, which allows the identification of subgroups with shared biological characteristics. Such stratification forms the cornerstone for developing actionable and personalized treatment strategies.

Researchers utilized multi-view clustering on genomic data to identify meaningful cancer subtypes, demonstrating that robust clustering techniques can reveal subtypes with distinct prognostic outcomes or therapeutic responses. For example, Subtype A, characterized by high immune cell infiltration, may respond better to immunotherapy, while Subtype B, marked by specific genetic mutations, could benefit from targeted therapies. By accurately stratifying patients, this approach enables clinicians to align treatments with the unique characteristics of each subtype, reducing trial-and-error in therapy selection and ultimately improving patient survival rates. Unsupervised clustering is highly suitable for multiomics cancer data analysis as it uncovers patterns without requiring predefined labels, enabling the discovery of novel cancer subtypes and biomarkers. Methods like KMeans and Hierarchical clustering allow researchers to explore hidden relationships within the data, revealing insights that supervised methods might miss. Unlike supervised clustering, which depends on labeled data and known outcomes, unsupervised approaches are more flexible and effective in identifying unknown subtypes, making them invaluable in the complex and dynamic field of cancer research.

A key enabler of this process is the integration of multiomics technologies, which provide a comprehensive view of the molecular underpinnings of diseases. By simultaneously analyzing multiple biological layers such as genomics, transcriptomics, proteomics, and epigenomics,

multiomics approaches have revolutionized disease subtype identification by leveraging the integrative analysis of diverse biological datasets. For instance, methods like NEMO (Speicher and Pfeifer, 2015) and PINSPlus (Nguyen et al., 2019) prioritize handling noise during clustering. PINSPlus, using Gaussian models, refines patient-patient similarity matrices to improve the robustness of subtype identification, while NEMO offers partial dataset utilization but lacks noise-handling capabilities. Dimensionality reduction-based approaches such as intNFM (Chalise and Fridley, 2017) and LRAcluster (Wu et al., 2015) rely on non-negative matrix factorization and low-rank approximations, respectively, to identify clusters using multiomic datasets. These methods assume linear associations across omic levels, which may oversimplify complex biological interactions. Similarly, CIMLR (Ramazzotti et al., 2018) employs Gaussian kernels for similarity computation, followed by dimensionality reduction and clustering. However, Gaussian-based methods are often sensitive to noise, limiting their effectiveness in real-world datasets.

On the other hand, statistical modeling approaches like iCluster (Shen et al., 2009), iClusterPlus (Mo et al., 2013), and MDI (Kirk et al., 2012) integrate multiomics data to identify cancer subtypes by constructing probabilistic models. These methods provide flexibility but often require careful parameter tuning. iCluF performs iterative fusion of multiomics datasets to identify clusters patients with cancer (Shakyawar et al., 2024). Deep-learning-based strategies, such as Subtype-GAN (Yang et al., 2021), leverage advanced architectures like the Gaussian Mixture model for subtype identification but demand larger sample sizes, making them less feasible for rare disease datasets. Integration-focused approaches such as similarity network fusion (SNF) (Wang et al., 2014) and PINS (Nguyen et al., 2017) aim to merge similarity matrices derived from individual omic layers before clustering. A similar study which helped in personalized treatment of cancer patients was "Cancer Subtype Identification based on Multi-view Subspace Clustering

with Adaptive Local Structure Learning" (Zhang et al., 2021). While these methods offer comprehensive integration, they are sensitive to noise and require balanced datasets for optimal performance.

Most studies leveraging resources such as ICGC (www.dcc.icgc.org) and The Cancer Genome Atlas (TCGA, www.genome.gov) focus on specific methods or predefined analytical pipelines, limiting flexibility and adaptability for broader research needs especially in the context of disease subtyping. At the same time, these approaches require programming skills and specific datasets in specific formats, which pose significant challenges for researchers to compare different methods and choose the best one.

To address these limitations, C3MOD was developed to provide a flexible platform for multiomics data analysis and cancer subtyping. We took TCGA methylation data, mRNA data and miRNA data to perform the analysis. Methylation data reveals key regulatory mechanisms in gene expression, with abnormal methylation often linked to cancer onset and progression (Chen et al., 2022). miRNA data highlights gene regulation processes, aiding in understanding metastasis and therapy resistance, while mRNA data reflects gene activity, revealing alterations critical for identifying cancer subtypes and therapeutic targets. Integrating these datasets enhances insights into cancer biology and prognosis. The workflow of C3MOD addresses the challenges of multiomics data complexity and integration, offering researchers a customizable and efficient framework for patient stratification, and further subtype-specific downstream analysis. In the present study, downstream analysis mainly focuses on survival analysis, mutation burden analysis, and clinical stage analysis of the patients in the identified subgroups. The overall strategy

facilitates the stratification of patients into actionable subgroups using the best modeling approach, ultimately supporting the development of precision medicine strategies.

2. Objectives

The overarching goal of this project is to develop an interactive and flexible platform, for analyzing multi-omics data from cancer patients, identify subtypes based on data integration and machine learning approach, and, finally, perform the downstream analysis including survival, and mutation burden characterization of the patients in the identified subtypes. The project unfolds in three phases, each comprising distinct specific aims, outlined as follows:

Specific aim 1- Implementation of multiple unsupervised clustering algorithms such as KMeans, SNF, Hierarchical, Spectral, FuzzyCMeans etc. to identify disease subtypes using TCGA's cancer multiomics data (level-3 data).

Specific aim 2- Characterization of identified subtypes (from specific aim 1) using biological and clinical information including survival analysis, mutation-based enrichment analysis, stage analysis and immune characterization.

Specific aim 3- Deployment of the tool/software on GitHub and provide comprehensive instructions and documentation to ensure user accessibility and usability.

3. Materials and Methods

3.1. Benchmark Datasets

A publicly available multi-omics datasets of 8572 patients with 30 cancer types was obtained from TCGA. The level-3 curated and normalized data for each cancer along with miRNA, mRNA, and DNA methylation dataset were downloaded from Xena (<https://xenabrowser.net/datapages/>). The omics datasets were preprocessed to remove “NA” in the data. The miRNA, mRNA, and CpG (methylation) with “zero” values for all patients were also removed from the dataset. Similarly, patients in any datasets with “zero values” throughout the columns were also removed. The TCGA cancer classification framework outline by Hoadley et al., 2018 was utilized to identify potential molecular features. These features were then compared with our processed dataset to find the common hits. The final sample information in each data type for each cancer type is provided in **Table 1**. The patient records that were present in all the three datasets: miRNA, mRNA and methylation were first identified. Then these datasets were combined into a single, unified patient matrix, which was used the clustering process. The data related to Gene mutations, Patients survival, and Clinical Stages (stage I, II, III, and IV) was downloaded from Xena. Immune score of TCGA patients was taken from Thorsson et al., 2018.

Table 1: Summary of patients data from TCGA that was used in the present study. The column “Common Samples” lists the number of common patients among miRNA, mRNA, and methylation datasets.

Sl. No.	TCGA abbreviation for cancer	Cancer type	Common Samples (miRNA, mRNA, methylation datasets)
1	TCGA-BRCA	Breast invasive carcinoma	849
2	TCGA-THCA	Thyroid carcinoma	562
3	TCGA-HNSC	Head and Neck squamous cell carcinoma	537
4	TCGA-PRAD	Prostate adenocarcinoma	533
5	TCGA-LGG	Brain Lower Grade Glioma	530
6	TCGA-LUAD	Lung adenocarcinoma	467
7	TCGA-UCEC	Uterine Corpus Endometrial Carcinoma	465
8	TCGA-BLCA	Bladder Urothelial Carcinoma	423
9	TCGA-LIHC	Liver hepatocellular carcinoma	411
10	TCGA-STAD	Stomach adenocarcinoma	370
11	TCGA-LUSC	Lung squamous cell carcinoma	368
12	TCGA-KIRC	Kidney renal clear cell carcinoma	346
13	TCGA-CESC	Cervical squamous cell carcinoma and endocervical adenocarcinoma	309
14	TCGA-COAD	Colon adenocarcinoma	307
15	TCGA-KIRP	Kidney renal papillary cell carcinoma	298
16	TCGA-SARC	Sarcoma	261
17	TCGA-ESCA	Esophageal carcinoma	195
18	TCGA-PCPG	Pheochromocytoma and Paraganglioma	187
19	TCGA-PAAD	Pancreatic adenocarcinoma	183
20	TCGA-TGCT	Testicular Germ Cell Tumors	156
21	TCGA-LAML	Acute Myeloid Leukemia	134
22	TCGA-THYM	Thymoma	122
23	TCGA-READ	Rectum adenocarcinoma	99
24	TCGA-MESO	Mesothelioma	87
25	TCGA-UVM	Uveal Melanoma	80
26	TCGA-ACC	Adrenocortical carcinoma	79
27	TCGA-KICH	Kidney Chromophobe	66
28	TCGA-UCS	Uterine Carcinosarcoma	57
29	TCGA-DLBC	Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	47
30	TCGA-CHOL	Cholangiocarcinoma	44
Total patients			8572

3.2. Cancer subtypes identification using unsupervised machine learning.

As mentioned in Specific Aim 1, multiple unsupervised clustering algorithms, including KMeans, SNF, Hierarchical, Spectral and FuzzyCMeans were used to identify patients' clusters. The miRNA, mRNA, and methylation data were concatenated (combined) to generate a final integrated matrix which was used as input to train the different models, followed by classification of the patients. WCSS (Within Cluster Sum of Squares) and Silhouette scores were used to recommend the best k (number of clusters) for each algorithm. Once clustering was completed, a different level of characterization of the identified subtypes was conducted, as explained in the next section. The overview of the workflow is provided in **Figure 1**.

3.3. Downstream analysis of identified subtypes

Survival analysis: Survival data of cancer patients were utilized to explore survival correlations within each identified subtype. To assess significant survival differences between the subtypes, Kaplan-Meier (KM) survival curves generated by the Cox proportional hazards (coxph) method was used. The statistical significance of the survival differences between each pair of identified clusters was determined using the log-rank test.

Mutation analysis: To assess whether the identified clusters contained patients with significant differences in their mutation profiles, the mutation profiles of all Differentially Expressed (DE) genes associated with patients in different subtypes were analyzed. The Chi-square test was used to determine the statistical significance of gene mutation differences between clusters, and the Pearson correlation matrix was applied to understand the similarity between pairs of clusters. Additionally, key information, such as the total number of mutations in each patient, the total number of mutations in each cluster, and the average number of mutations per patient in each identified cluster were summarized. To enhance visual clarity and interpretability of the data,

a variety of visualizations were generated, such as bar graphs, Pearson correlation matrices, and density plots.

Stage analysis: A stage analysis was conducted to examine the clinical stage distribution of patients across the different identified clusters. This analysis provides valuable insights into the distribution of patients at various stages within each cluster and supports the hypothesis that survival outcomes differ among the identified clusters, which can inform the development of tailored treatment strategies for patients in each cluster.

Immune characterization: The previous study titled “The Immune Landscape of Cancer” (Thorsson et al., 2018) conducted an extensive immunogenomic analysis of over 10,000 tumors from 33 different cancer types, using data from TCGA. It identified six distinct immune subtypes and calculated immune scores for 160 gene signatures. In our current analysis, these immune scores were used to perform a Pearson correlation test to evaluate the similarity between different pairs of the identified clusters.

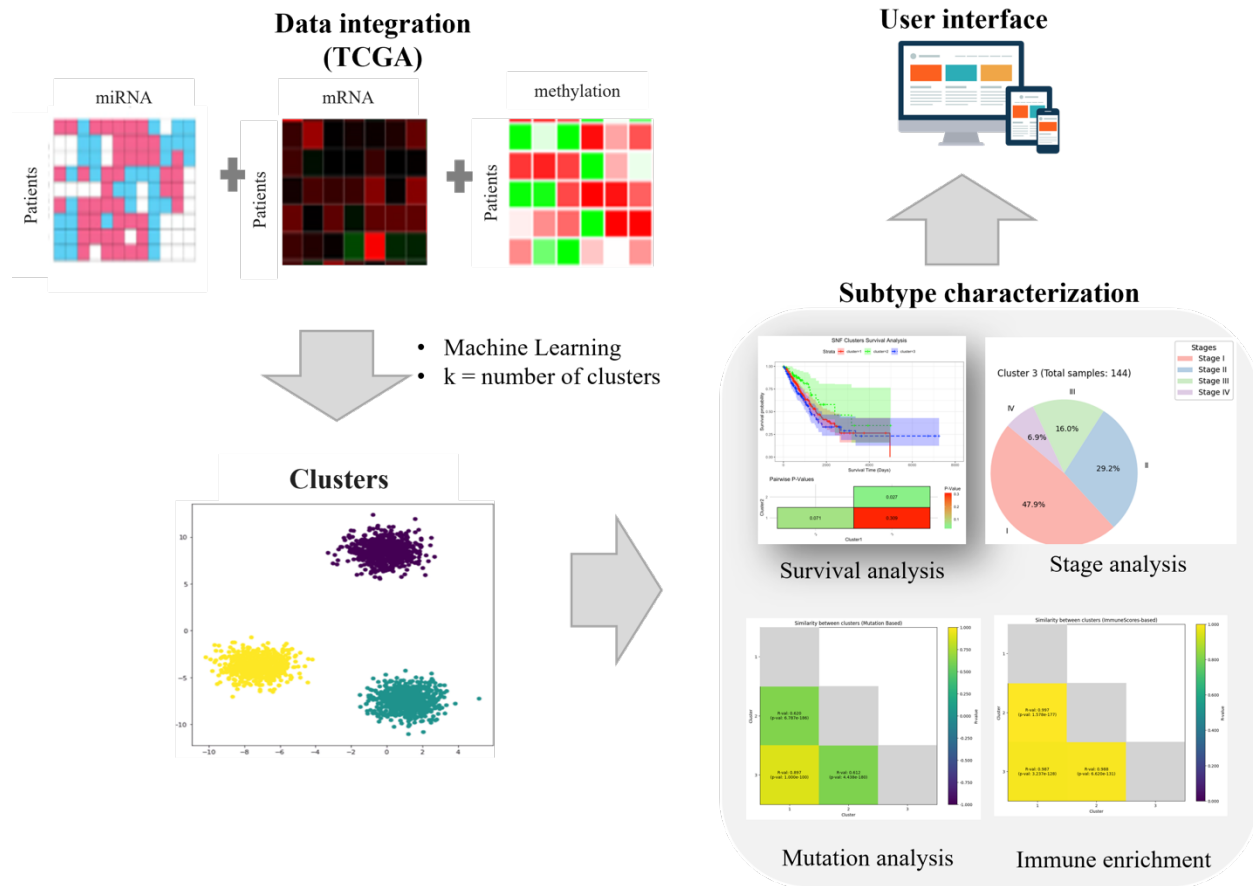


Figure 1: The overview of the C3MOD workflow including data integration, cluster identification, and downstream characterization.

3.4. Software Development

The development of the C3MOD tool involves creating multiple Python scripts, each designed to perform a specific task, including library installation, data preprocessing, recommendation of k (number of clusters), clustering, and post-clustering analysis. These scripts run sequentially based on user inputs, forming a pipeline. Results are stored in an output folder at each stage. The workflow is illustrated in **Figure 1**.

Libraries and Packages Used:

Python Libraries: NumPy 1.23.4, Pandas 1.5.3, scikit-learn 1.2.0, rpy2 3.5.8, matplotlib 3.6.2, skfuzzy 0.4.2, seaborn 0.12.2, pickle 5.0.0

R Libraries: SNFtool 2.2.0, data.table 1.14.8, dplyr 1.1.0, ggplot2 3.4.2, factoextra 1.0.7, cluster 2.1.3, survival 3.4-0, survminer 0.4.9, pryr 0.1.4, bigmemory 1.6.2, ggtext 0.1.1, cowplot 1.1.1.

The scripts development related workflow is provided in the below box “Code development workflow”

Code development workflow-

Step 1: Data Preprocessing

- Data Cleaning: Handle missing values, duplicates, and erroneous data.
- Sample Integration: Identify common samples across miRNA, mRNA, and methylation data for integration.
- Standardization: Normalize data to ensure uniformity across features.

Step 2: Recommendation of k (number of clusters)

- WCSS Plot: Identify optimal K using the "elbow" method.
- Silhouette Score: Assess cluster quality for optimal K.

Step 3: Clustering Algorithms

- SNF, KMeans, Hierarchical, Spectral, FuzzyCMeans
- PCA for cluster visualization.

Step 4: Post-Clustering Analysis

- Survival Analysis: CoxPH, Log-rank test, and Kaplan-Meier curve.
- Mutation Analysis: Chi-square, Pearson correlation, and visualization.
- Stage Analysis: Data visualization of cancer stages.
- Immune Characterization: Pearson correlation and visualization.

Scripts run in a defined sequence based on user inputs, with results saved in an output folder at each stage.

Deployment for public access

The source code has been pushed to a public repository on [GitHub – C3MOD](#) for public access. Detailed, step-by-step documentation along with a demonstration video is available in the README file. Anyone can clone the repository and, by following the simple steps outlined in the README file, can perform customized subtyping and analysis based on their specific research interests.

4. Results and discussion

A case study using TCGA-LUAD (Lung adenocarcinoma cancer dataset)

To demonstrate the usability of C3MOD, a case study was performed using TCGA-LUAD dataset (Total patients = 467), and downstream subtype-level characterization.

4.1. Data pre-processing and recommendation for k (number of clusters)

All the data preprocessing and integration steps were conducted on the miRNA, mRNA, and methylation data. Additionally, the WCSS (Within-Cluster Sum of Squares) (**Figure 2B**) and Silhouette plot (**Figure 2A**) were generated to help choose the most suitable value for **k**. The highest value of the Silhouette score indicates that $k=2$ is the most suitable number of clusters, while the elbow point of the WCSS curve suggests that $k=3$ may also be a viable option.

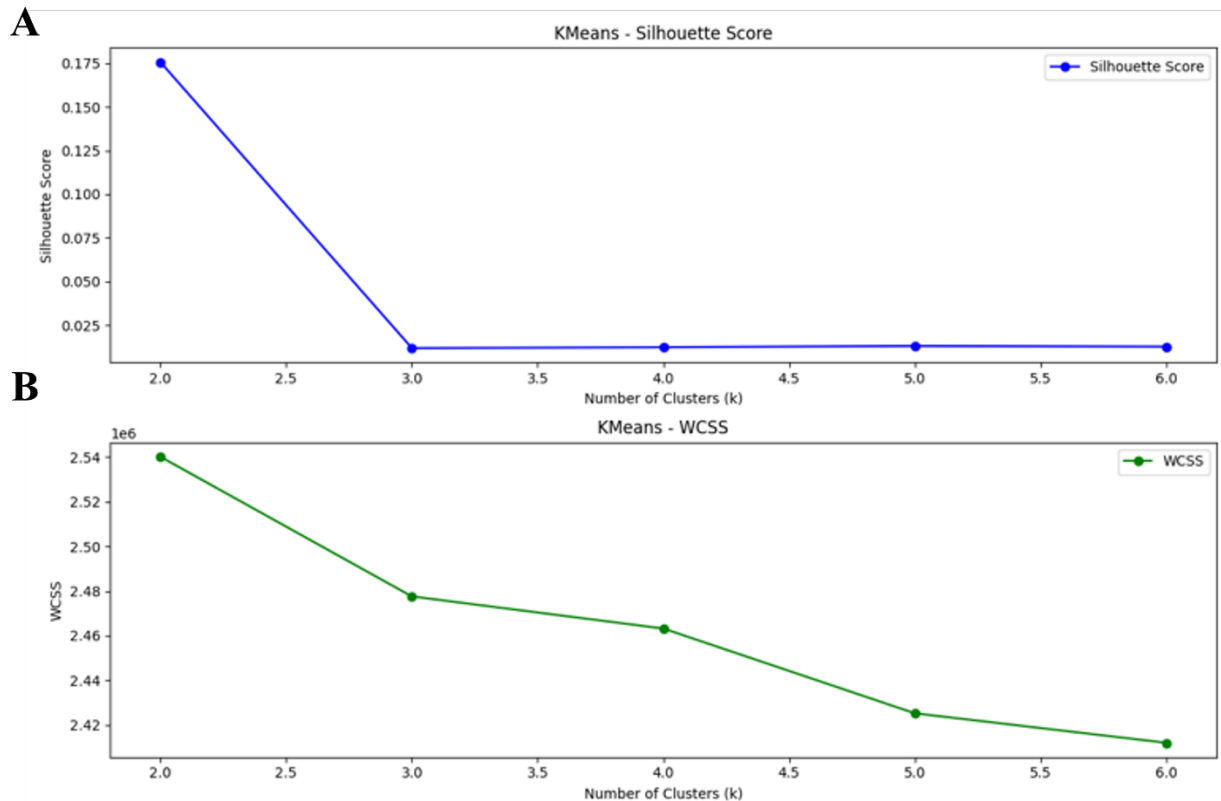


Figure 2: Illustration of WCSS and Silhouette plots for the KMeans algorithm using TCGA-LUAD dataset to recommend number of clusters (k).

4.2. Subtype identification

Using 467 common patients among miRNA, mRNA and methylation dataset as input, three clusters were identified (at $k=3$). Principal Component Analysis was performed on the output clusters and visualized in scatter plots, as shown in **Figure 3**. Cluster 1, 2, and 3 include 267, 43, and 132 patients, respectively.

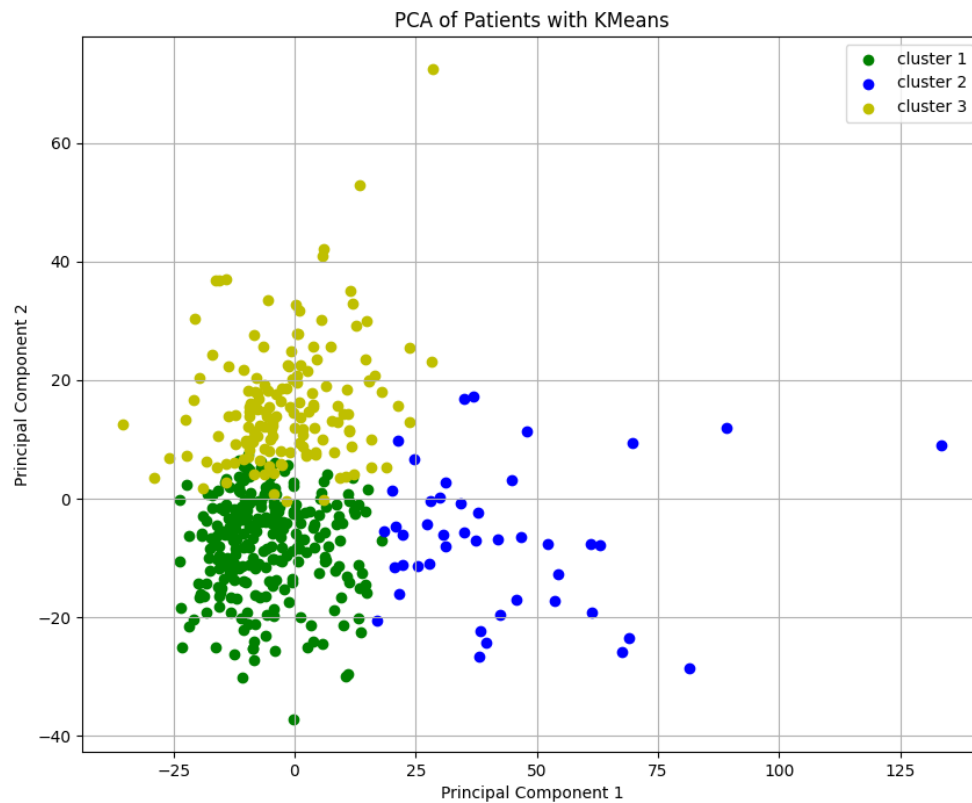


Figure 3: PCA plot illustrating three separate clusters by KMeans algorithm on TCGA-LUAD dataset.

4.3. Survival analysis of three identified clusters

Survival analysis was conducted to investigate the three identified clusters using the Cox proportional hazards (coxph) method. A survival probability vs. survival time plot was created to visualize the outcomes. **Figure 4** presents a matrix of p-values comparing clusters, highlighting the statistical significance of differences. The results indicate that for LUAD, clusters 1 and 3 show a significant difference in survival profiles (p-value = 0.023), while other pairings are not significant. Patients in cluster 3 exhibited the highest survival rates, whereas those in cluster 1 had the lowest.

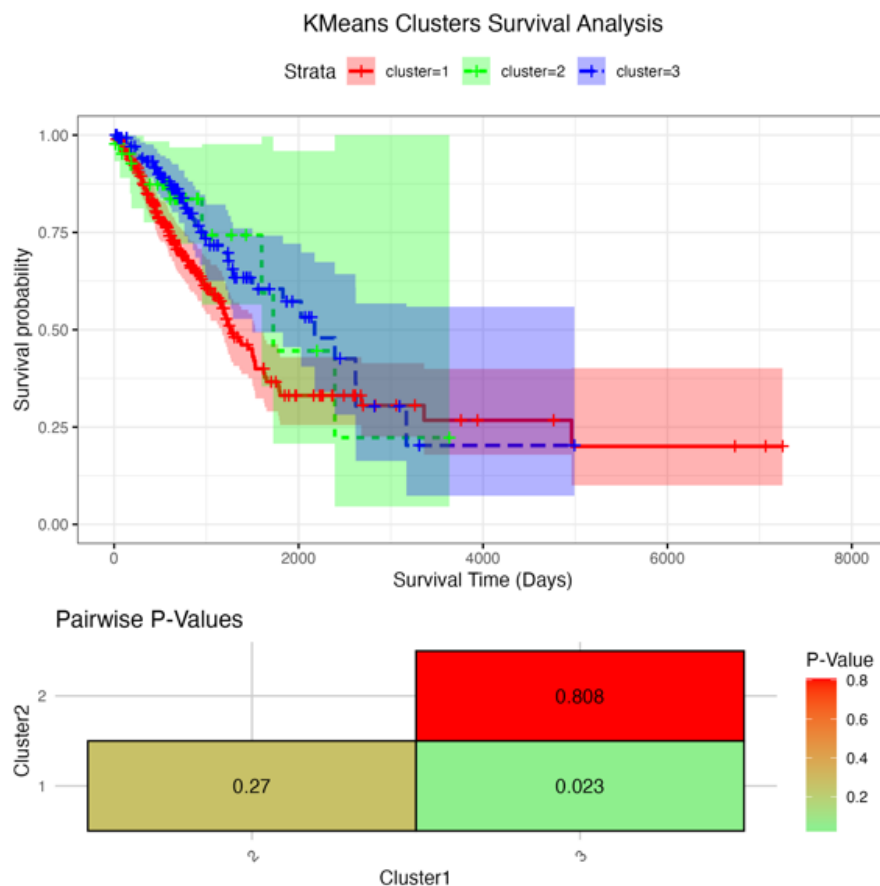


Figure 4: Kaplan-Meier (KM) survival curves and survival correlation of different subtypes by KMeans algorithm using TCGA-LUAD dataset.

4.4. Stage analysis of identified subtypes

Stage analysis was conducted to visualize the clinical stage distribution among patients of different clusters. Pie charts were drawn to illustrate the percentage of patients in different clinical stages across clusters, as shown in **Figure 5**. The results indicate that in LUAD patients, Cluster 1 and Cluster 3 exhibit a higher proportion of early-stage patients, with 51.3% and 62.9% , respectively, being diagnosed with Stage I. In contrast, Cluster 2 has a greater number of patients in later stages, with 46.5% being diagnosed with Stage III. This highlights distinct clinical stage distributions across the clusters, suggesting potential differences in disease progression.

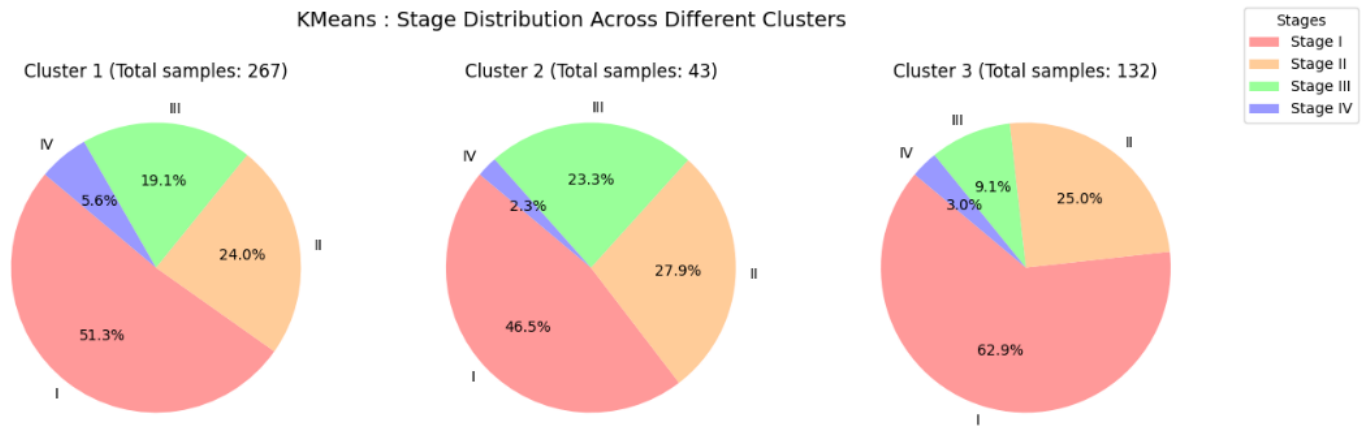


Figure 5: Stage Distribution of patients across different clusters identified by KMeans algorithm using TCGA-LUAD dataset.

4.5. Mutation correlations of disease subtypes

Mutation analysis was performed to conduct statistical tests on each gene across all cluster pairs. In addition to generating mutation summaries and visualizations, a Pearson correlation matrix was constructed to assess the similarity between the identified clusters. The results indicate that genes with significant mutational differences were identified in each pair-wise comparison: HAS1 (p-value < 0.005), GNG4 (p-value < 0.004), and TRPMS (p-value < 0.005) for cluster pairs 1 vs. 21 vs. 3, and 2 vs. 3, respectively (**Figure 6A-C**). Notably, the most significant genes identified were. Furthermore, aggregated mutational differences showed a high similarity between clusters 1 and 3 ($R = 0.81$, p-value = $1.00\text{e-}100$) as illustrated in **Figure 6D**.

The gene distribution analysis revealed, that only a few genes (highlighted in red bars) were found to be significant across clusters, as shown in **Figures 7A-C**. Additional analysis revealed that cluster 1 exhibited the highest number of gene mutations, while cluster 2 had the lowest (**Figure 7D**). Interestingly, the average number of mutations per patient was highest in cluster 2 and lowest in cluster 3 (**Figure 7E**). Notably, the LUAD dataset predominantly comprises patients with relatively fewer mutations (**Figure 7F**).

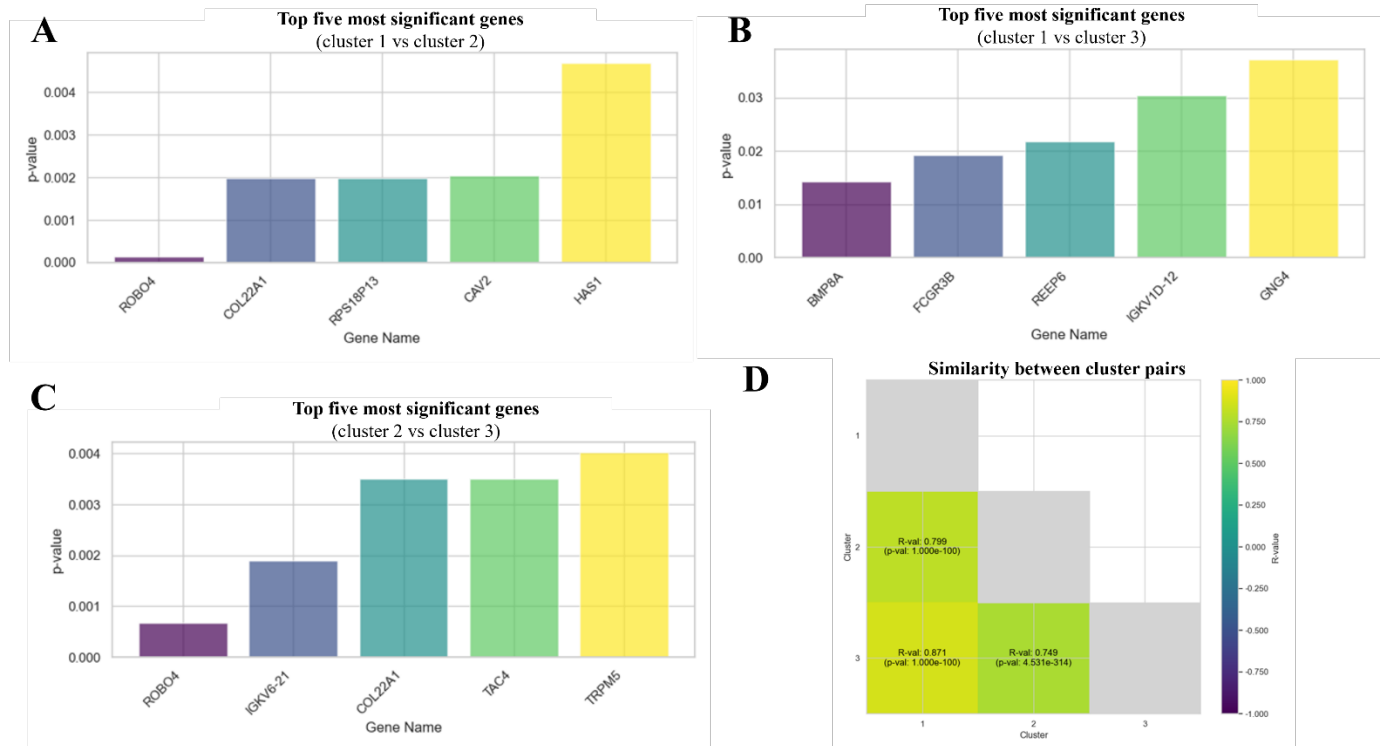


Figure 6: Top five genes showing maximum significant mutational difference across clusters 1 and 2 (A), clusters 1 and 3 (B), and clusters 2 and 3 (C). D) Similarity matrix showing mutation-based correlation between different pair of clusters.

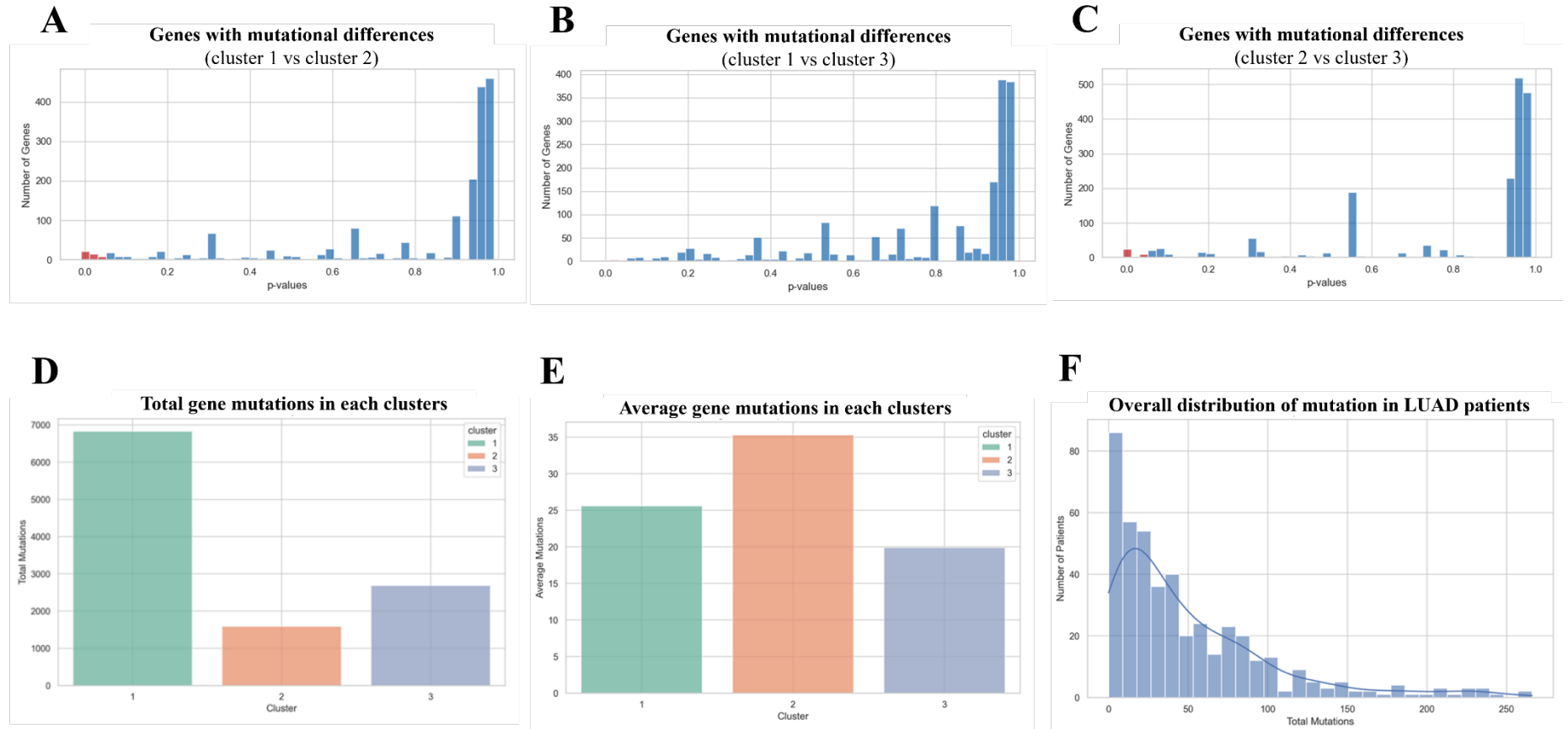


Figure 7: Distribution of genes with different calculated p-values (significance of mutational differences) between clusters 1 and 2 (A), clusters 1 and 3 (B), and clusters 2 and 3 (C). Distribution of total gene mutations (D) and average mutations (E) in patients with different clusters are shown. The overall distributions of mutations across patients are shown in (F).

4.6. Immune Characterization of identified subtypes

Immune characterization was performed using the immune scores of 160 gene signatures, as described previously in the methods section. The results show high correlations among all clusters, with R-value (**Figure 8**) indicating the strength of correlations and p-values demonstrating their statistical significance.

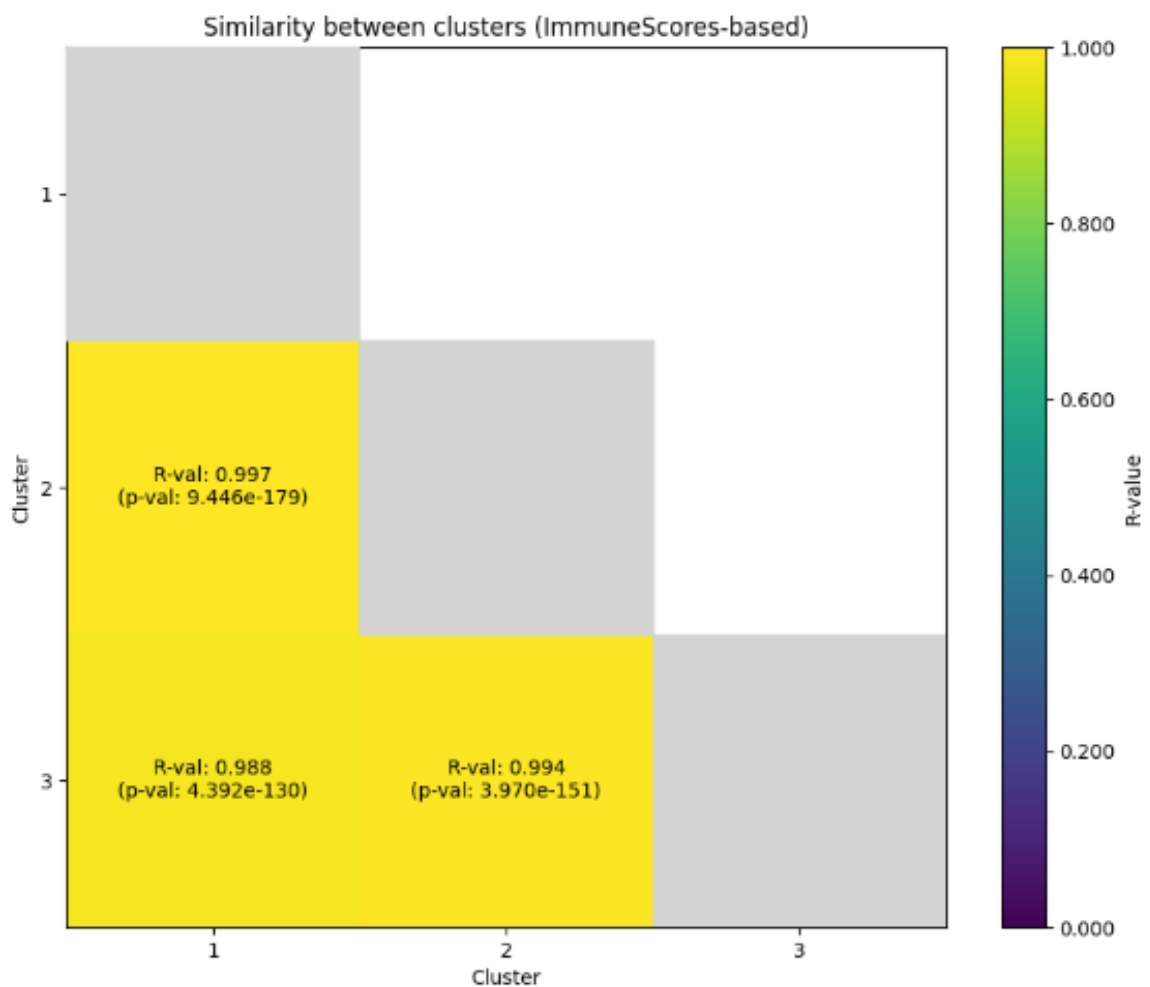


Figure 8: Similarity matrix showing Immune score-based correlation between different pairs of clusters.

4.7. Identification of gene **GNG4** as a potential biomarker

The mutation analysis performed on the identified subtypes using the KMeans and SNF algorithms on TCGA-LUAD data revealed valuable insights regarding the gene **GNG4**, which plays a key role in LUAD cancer. Studies have shown its significant correlation with poor survival outcomes in lung cancer patients, particularly in lung adenocarcinoma (LUAD) (Jiang et al, 2023). This gene has been identified as an important biomarker, with its overexpression being associated with tumor progression and worse prognosis (Wen et al., 2022).

The **C3MOD** tool also identifies statistically significant differences in **GNG4** expression between clusters found by the KMeans algorithm (between clusters 1 and 3) and the SNF algorithm (between clusters 2 and 3), suggesting that **GNG4** may influence the survival outcomes of these specific clusters. The Cox proportional hazards (coxph) survival outcome plots generated by the tool for KMeans (k=3) in **Figure 9** and SNF clustering algorithms in **Figure 10** provide further validation of the statistically different survival outcomes between these pairs of clusters, reinforcing the potential of **GNG4** as a critical player in lung cancer survival prognosis.

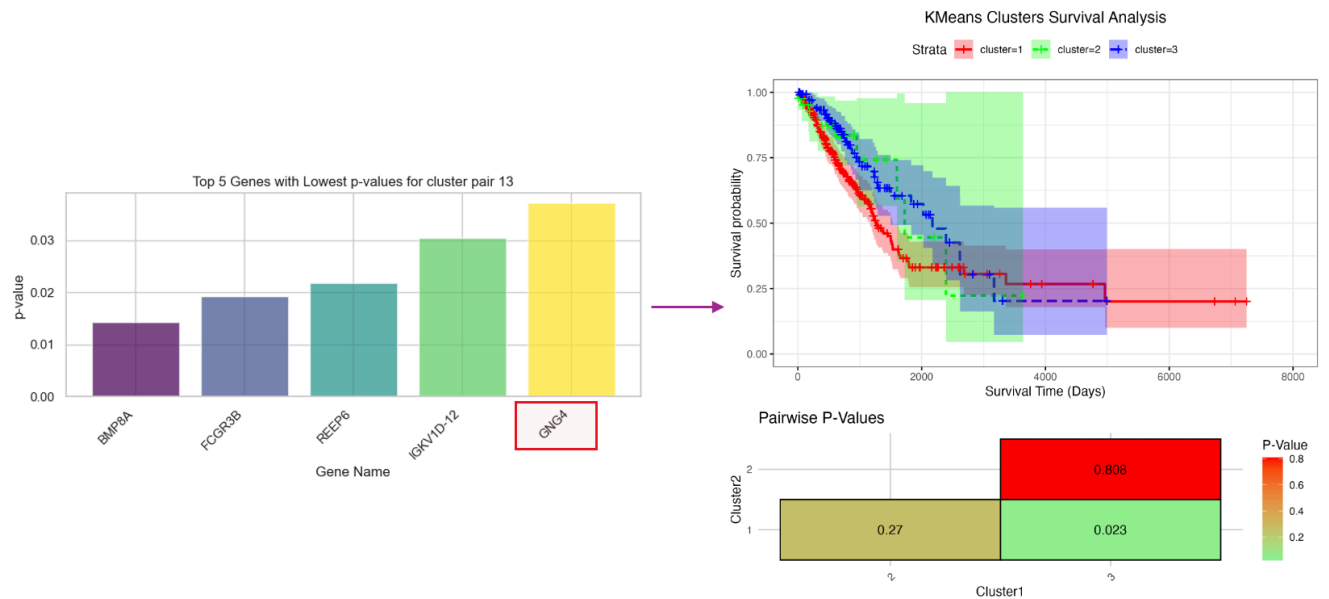


Figure 9: GNG4 as a most significant gene with highest mutational difference in patients in cluster 1 and 3 (clustering performed using KMeans algorithm). These clusters show significant differences in survival profile of their patients.

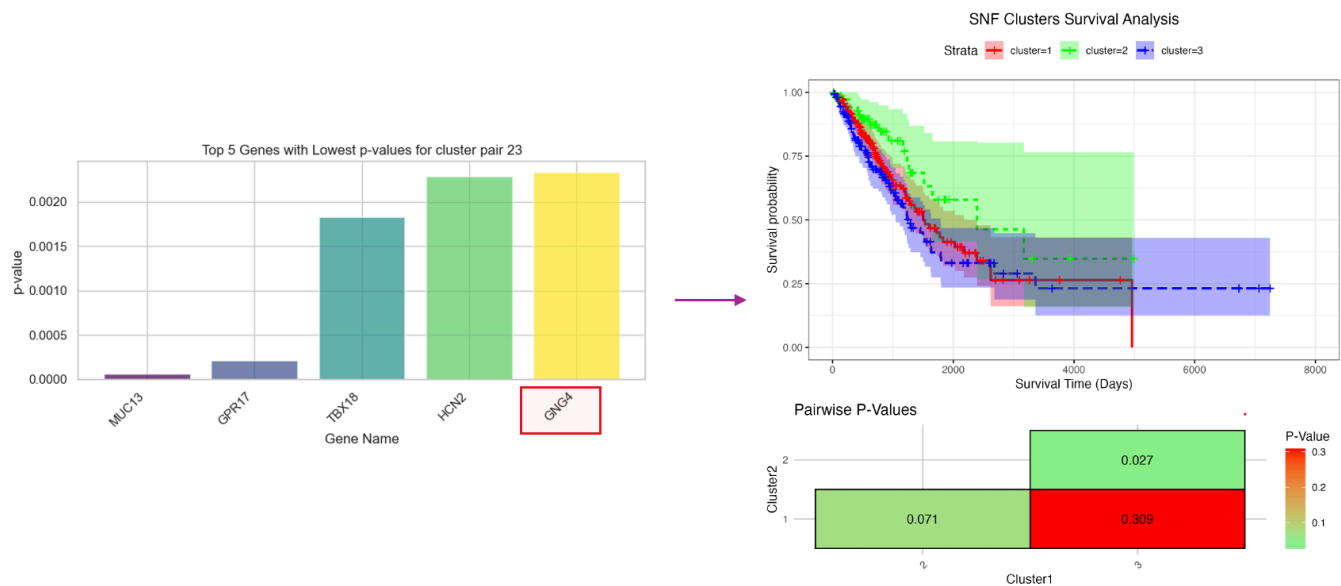


Figure 10: Identification of GNG4 as a most significant gene with highest mutational difference in patients in cluster 2 and 3 (clustering performed using SNF algorithm). These clusters show significant differences in survival profile of their patients.

5. Conclusions

The C3MOD tool represents a significant advancement in cancer research. It provides valuable insights into the molecular and clinical heterogeneity of cancer. By integrating multiomics data from various biological layers—miRNA, mRNA, and methylation—C3MOD offers a comprehensive framework for patient stratification, enabling precision medicine strategies tailored to specific cancer subtypes. Its implementation of multiple unsupervised clustering algorithms, robust downstream analyses such as survival, mutation, and immune characterization, and its user-centric deployment on GitHub make it an invaluable resource for researchers.

The successful identification of clinically significant biomarkers like GNG4 in lung adenocarcinoma demonstrates C3MOD's potential to uncover actionable insights that improve cancer prognosis and therapeutic interventions. Furthermore, C3MOD contributes significantly to the advancement of cancer research by providing innovative analytical capabilities to (1) Identify key genes and biochemical pathways (2) visualize cluster similarities (3) analyze stage distribution across clusters and (4) provide immune analysis and personalized treatment strategies. Some specific examples that demonstrated the potential of C3MOD to facilitate a deeper understanding of cancer biology and develop more effective treatment strategies include:

- 1) **Key Gene Identification and Pathway Analysis:** Our analysis using C3MOD identified key genes associated with specific cancer subtypes, such as BMP8A, FCGR3B, and REEP6, which showed strong correlations between clusters 1 and 3. These genes were found to be enriched in pathways related to cell migration and invasion, suggesting a crucial role in tumorigenesis.
- 2) **Visualizing Cluster Similarity:** C3MOD generated a correlation matrix showing strong correlations between clusters 1 and 3 ($R\text{-value} = 0.871$, $p\text{-value} < 1e-100$), indicating

similar mutation patterns. This information was further visualized using a heatmap, which displayed the similarity in mutation profiles across all clusters. These results enable researchers to identify potential relationships between different subtypes.

- 3) **Analyzing Stage Distribution Across Clusters:** Our analysis revealed significant differences in stage distribution across clusters ($p\text{-value} = 0.0001$), suggesting that certain subtypes may be more aggressive or responsive to treatment. This information can be used to develop targeted treatment approaches, such as more aggressive therapy for high-stage tumors.
- 4) **Immune Analysis and Personalized Treatment Strategies:** C3MOD calculated an immune score (e.g., CD8+ T-cell infiltration) for each cluster, revealing significant differences in immune cell populations across subtypes. The analysis suggested that certain subtypes may benefit from targeted therapies such as CD8+ T- cells.

By providing advanced analytics capabilities, C3MOD empowers researchers and is poised to become an essential tool in the research community arsenal for advancing our understanding the complexities of cancer biology and improving patient outcomes.

6. Software and code availability

The C3MOD is a novel software application designed to empower cancer researchers in deciphering complex genomic landscapes. The tool is designed to be user-friendly and caters to researchers, clinicians, and biotech professionals in oncology. It is expected to serve as a valuable purpose by aiding in biomarker discovery, targeted therapy development, and exploration of novel treatment approaches. To foster collaboration and advancements within the

cancer research community, this tool is made available through open-source platform like the GitHub and can be accessed for download (<https://github.com/rajulshakya777/C3MOD>).

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